

Nivritha Polaboina

+91 9550564952 — nivritha.pola@gmail.com — GitHub — LinkedIn — Portfolio

SUMMARY

Computer Science undergraduate with strong experience in **Full-Stack Development, Machine Learning, NLP, and Data Analytics**. Proven ability to build scalable applications, optimize ML models, and develop real-world AI solutions. Experienced across the software development lifecycle, from data preprocessing to deployment.

EDUCATION

Anurag University, Hyderabad B.Tech in Computer Science Engineering	2023 – 2027 CGPA: 8.60
Trividhya Junior College, Hyderabad Higher Secondary	2021 – 2023 CGPA: 9.48
Amrita Vidyalayam, Hyderabad Secondary School	2011 – 2021 Percentage: 89

EXPERIENCE

Machine Learning Intern Skillfied Mentor, Hyderabad, Telangana	June 2025 – July 2025
- Built and trained ML models on real-world datasets using Python - Achieved improved prediction accuracy by 12–18% through optimization	
MERN Stack Intern Webstack Academy, Hyderabad, Telangana	Dec 2025 – Jan 2026
- Developed full-stack web applications using the MERN stack - Improved application performance and user experience by 20–30%	
Data Analytics Virtual Intern — Deloitte Australia (Forage) Virtual Experience Program	June 2025
Completed a virtual job simulation focused on data visualization, data cleaning, and business insights. Created Tableau dashboards from Excel data, delivering 3+ actionable insights for business decision-making	

PROJECTS

Collab Sphere: Digital Whiteboard Spring Boot, React, WebSocket	<i>Live Demo</i>
- Developed a real-time collaborative platform enabling multiple users to draw and interact simultaneously - Implemented WebSocket-based low-latency communication for instant synchronization - Designed scalable backend architecture to handle concurrency and session state management - Built features including drawing tools, object tracking, and session persistence	
Synthesia Music Player MERN Stack (MongoDB, Express, React, Node.js)	<i>Live Demo</i>
- Built a full-stack music streaming application with secure authentication and authorization - Implemented mood-based playlist recommendations and favorites system - Designed a custom audio player with responsive UI and optimized user experience - Developed RESTful APIs and improved backend performance through efficient data handling	
VibeSync: Emotion-Based Music Recommendation System React, Vite, Tailwind CSS, Flask, DeepFace, TensorFlow, MongoDB Atlas, Jamendo API	<i>Live Demo</i>
- Developed an AI-powered web application that detects user emotions in real time through webcam-based facial expression analysis - Integrated DeepFace and TensorFlow models for high-accuracy emotion detection and mood classification	

- Implemented personalized music recommendations by mapping emotions to curated genres and mood-based tracks
- Built secure JWT-based authentication and integrated Jamendo API for seamless music streaming experience
- Designed a responsive and immersive UI using React and Tailwind CSS with optimized frontend-backend communication

PATENTS

- **Patent Filed:** “AI Virtual Professor”
AI-powered adaptive virtual teaching and personalized learning system.

SKILLS

- **Programming Languages:** Python, Java, C, C++, JavaScript, TypeScript, R
- **Web & Full-Stack Development:** HTML, CSS, JavaScript, TypeScript, MERN Stack (MongoDB, Express.js, React.js, Node.js), Next.js, REST APIs, Bootstrap, Tailwind CSS, Responsive Web Design
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Spring Boot, Hibernate/JPA, Prisma ORM, TanStack Query, Zustand, React Hook Form, Zod, Framer Motion, Flask, FastAPI, PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
- **Artificial Intelligence & Machine Learning:** Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, NLP, OpenAI API Integration, Google Gemini API, AI Workflow Automation
- **Data Science & Analytics:** NumPy, Pandas, Matplotlib, Seaborn, OpenCV, NLTK, spaCy, Data Analysis, Data Visualization, Statistical Analysis
- **Databases:** MongoDB, PostgreSQL, MySQL, Neon PostgreSQL
- **Authentication & Security:** OAuth 2.0, Google Authentication, NextAuth.js, JWT Authentication
- **Cloud, DevOps & Deployment:** AWS Cloud Practitioner Essentials, Vercel, Docker, Linux, CI/CD, Environment Configuration, GitHub Actions
- **Tools & Platforms:** Git, GitHub, VS Code, Postman, Jupyter Notebook, Spring Tool Suite (STS), Tableau, Power BI, Qlik
- **Software Engineering Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Development Life Cycle (SDLC)

ACHIEVEMENTS

- **Winner — National-Level Hackathon:** Secured 1st place among 300+ participants by developing KPI-driven Qlik Cloud dashboards to analyze urban vehicle emissions and provide actionable insights
- **Runner-Up — Ideathon:** Achieved 2nd place for presenting an innovative and scalable solution to a real-world problem
- **Hackathon Finalist:** Reached advanced rounds in Microsoft x Reskill and Adobe India Hackathons, including Top 9 in a GenAI Hackathon
- **Selected — OpenAI Academy x NxtWave Buildathon:** Chosen at state level for innovation and technical problem-solving skills
- **Selected — Graphs Camp:** Chosen among 80,000+ applicants in a competitive selection process led by Codeforces Master
- **Certifications:** Certified in Python, C, SQL, NLP, Frontend Development (Infosys, Udemy, Cisco); AI, ML, Generative AI (Microsoft, MathWorks); MongoDB, Qlik, Networking Essentials (Cisco)

LEADERSHIP

- **Head of Public Relations — MALAI (Machine Learning & AI Club):** Led publicity and outreach for 6+ technical events including hackathons, workshops, and tech talks; increased participation to 200+ students and strengthened industry collaborations
- **Technical Team Member — ACM Student Chapter:** Organized and hosted technical events, coding sessions, and panel discussions; contributed to fostering a collaborative learning environment